

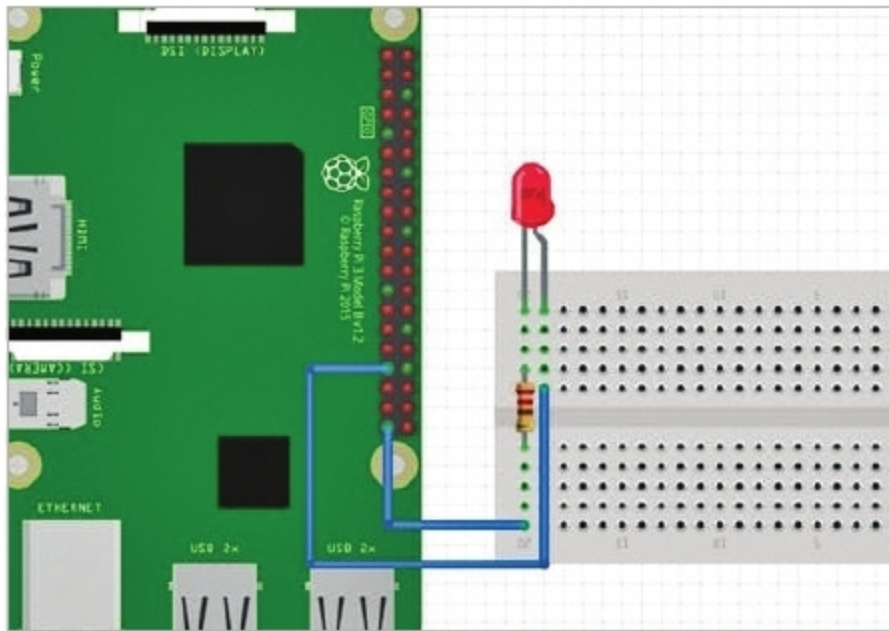


Figure 9: The design of circuit layout

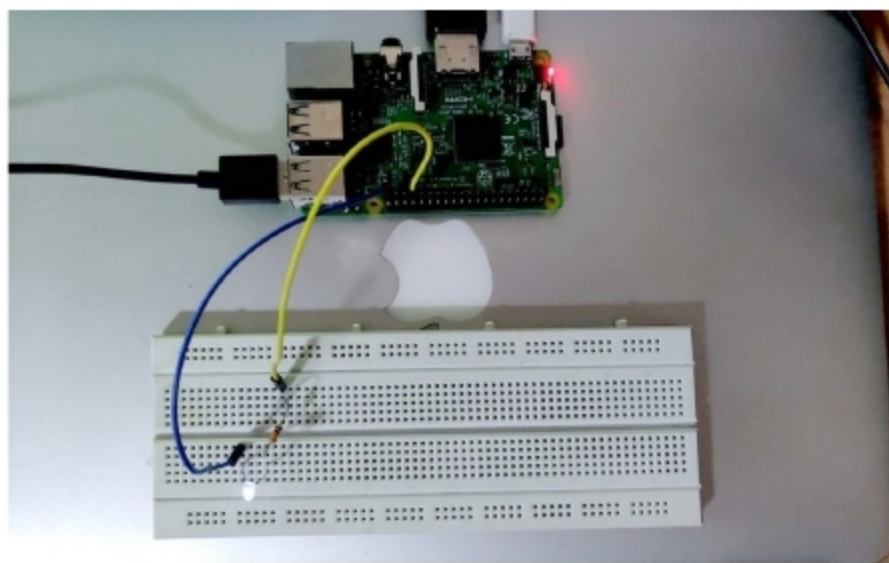


Figure 10: Final output

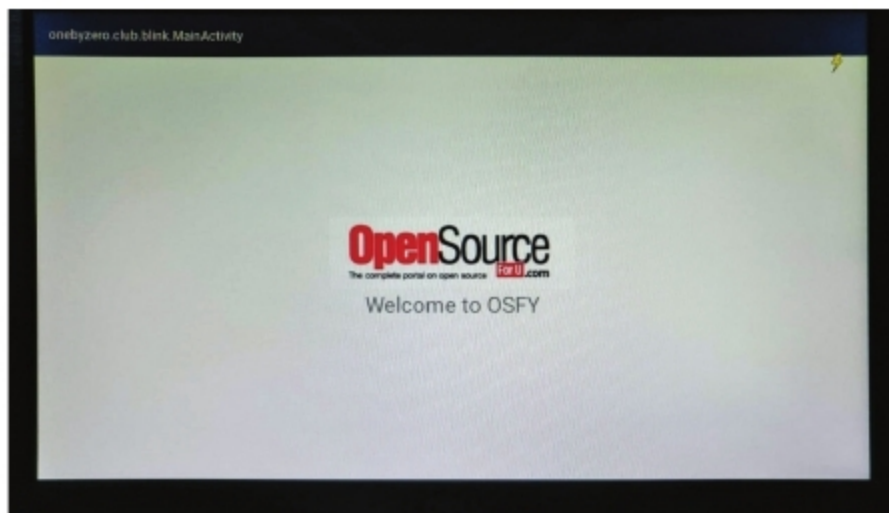


Figure 11: Redesigned layout for home page

```

DEVICE_RPI3 -> "BCM13"
else -> throw IllegalStateException("Unknown Build.DEVICE
${Build.DEVICE}")

}

```

Declare the following variables in the main class:

```

private val handler = Handler()
private lateinit var ledGpio: Gpio
private var ledState = false

```

In order to start the flashing, a handler has to be initialised. So inside the *onCreate* function, add the following command:

```

ledGpio = PeripheralManager.getInstance().
openGpio(BoardDefaults.gpioForLED)
ledGpio.setDirection(Gpio.DIRECTION_OUT_INITIALLY_LOW)
handler.post(blinkRunnable)

```

The above code will connect the GPIO and set the initial output as LOW. Also, the handler is called. The code for the handler is as follows:

```

private val blinkRunnable = object : Runnable {
    override fun run() {
        // Toggle the GPIO state
        ledState = !ledState
        ledGpio.value = ledState
        Log.d(TAG, "State set to ${ledState}")

        // Reschedule the same runnable in {#INTERVAL_
        BETWEEN_BLINKS_MS} milliseconds
        handler.postDelayed(this, INTERVAL_BETWEEN_
        BLINKS_MS)
    }
}

```

```

companion object {
    private const val INTERVAL_BETWEEN_BLINKS_MS = 100L
}

```

The handler needs to be closed on the application's exit, and the pins initialised must be freed. For that, add the following code to the *onDestroy* function:

```

override fun onDestroy() {
    super.onDestroy()
    handler.removeCallbacks(blinkRunnable)
    ledGpio.close()
}

```

Android Things and Android Studio are connected using Android Wifi ADB Connect (Figure 8).

To set up the circuit, use the schematic given in Figure 9. Switch on the Pi board and run the project on Android Studio. The LED will start blinking when the APK has finished installing (Figure 10).

Like when designing on Android, we can also add custom designs on Android Things, for which you can edit the resource layout file using Android Studio (Figure 11). **END** 🐧

By: Jis Joe Mathew

The author is an assistant professor of Computer Science and Engineering at the Amal Jyothi College of Engineering, Kanjirapally, Kerala. He can be contacted at jisjoemathew@gmail.com.